

SUPERFETCH & READYBOOST

Windows Vista Memory Management

Dave Coons

Vista's Improvements

- ▣ Windows Vista was released to the public on January 30th, 2007.
- ▣ Some one of its many new features were Superfetch and ReadyBoost
 - Improvement on virtual memory management
 - Designed to improve responsiveness
 - ▣ Different from performance
 - Responsiveness is how long it takes to start something
 - Performance is how long it takes to finish

Vista Memory Management

- ▣ Vista introduced two new memory management features
 - Superfetch: For machines with plenty of memory
 - Readyboost: For machines with minimal memory

Superfetch

- ▣ An improvement over Windows XP prefetch
- ▣ Monitors computer activity and retrieves page usage histories
- ▣ Directs the Memory Manager to preload data and code from files on disk or from a paging file.
 - Looks for patterns and attempts to preload pages that might be needed later.

Superfetch

- ▣ Whenever memory becomes, SuperFetch asks the Memory Manager to fetch data and code that was recently evicted
- ▣ This is done at a rate of a few pages per second with Very Low priority I/Os
 - This way the preloading does not impact the user or other active applications

▣ 1:<http://technet.microsoft.com/en-us/magazine/2007.03.vistakernel.aspx>

Superfetch

- ❑ Essentially extends page-tracking to data and code that was once in memory, but that the Memory Manager has reused to make room for new data and code
- ❑ It stores this information in scenario files with a .db extension in the %SystemRoot%\Prefetch directory alongside standard prefetch files used to optimize application launch
- ❑ Using this deep knowledge of memory usage, SuperFetch can preload data and code when physical memory becomes available

❑ 1:<http://technet.microsoft.com/en-us/magazine/2007.03.vistakernel.aspx>

Readyboost

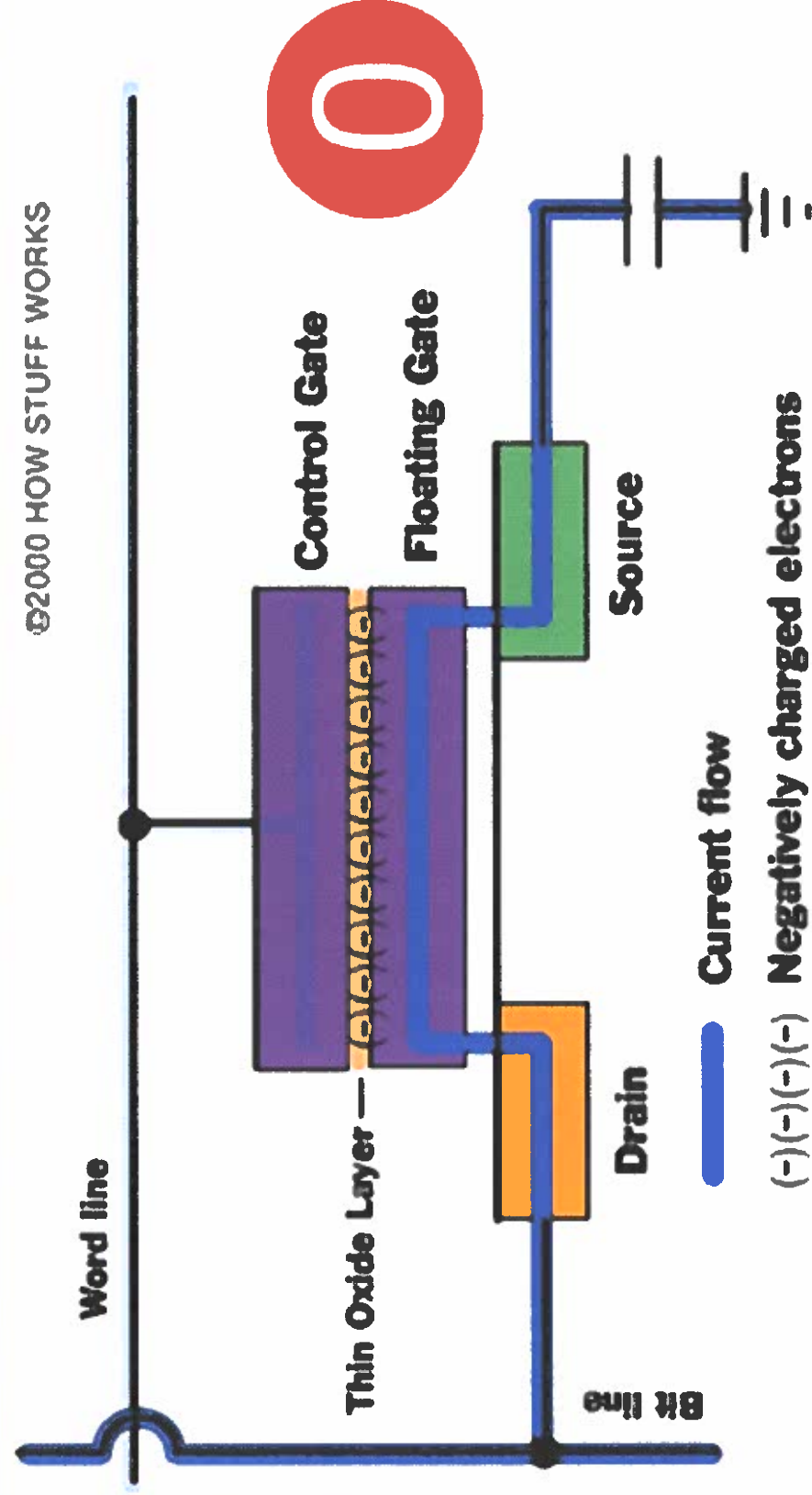
- ▣ Vista's minimum memory requirement is 512k
- ▣ A bit like a fat girl in a tube dress, It will work, but it ain't pretty
- ▣ Readyboost can help improve performance by using flash memory as another option for the backing store



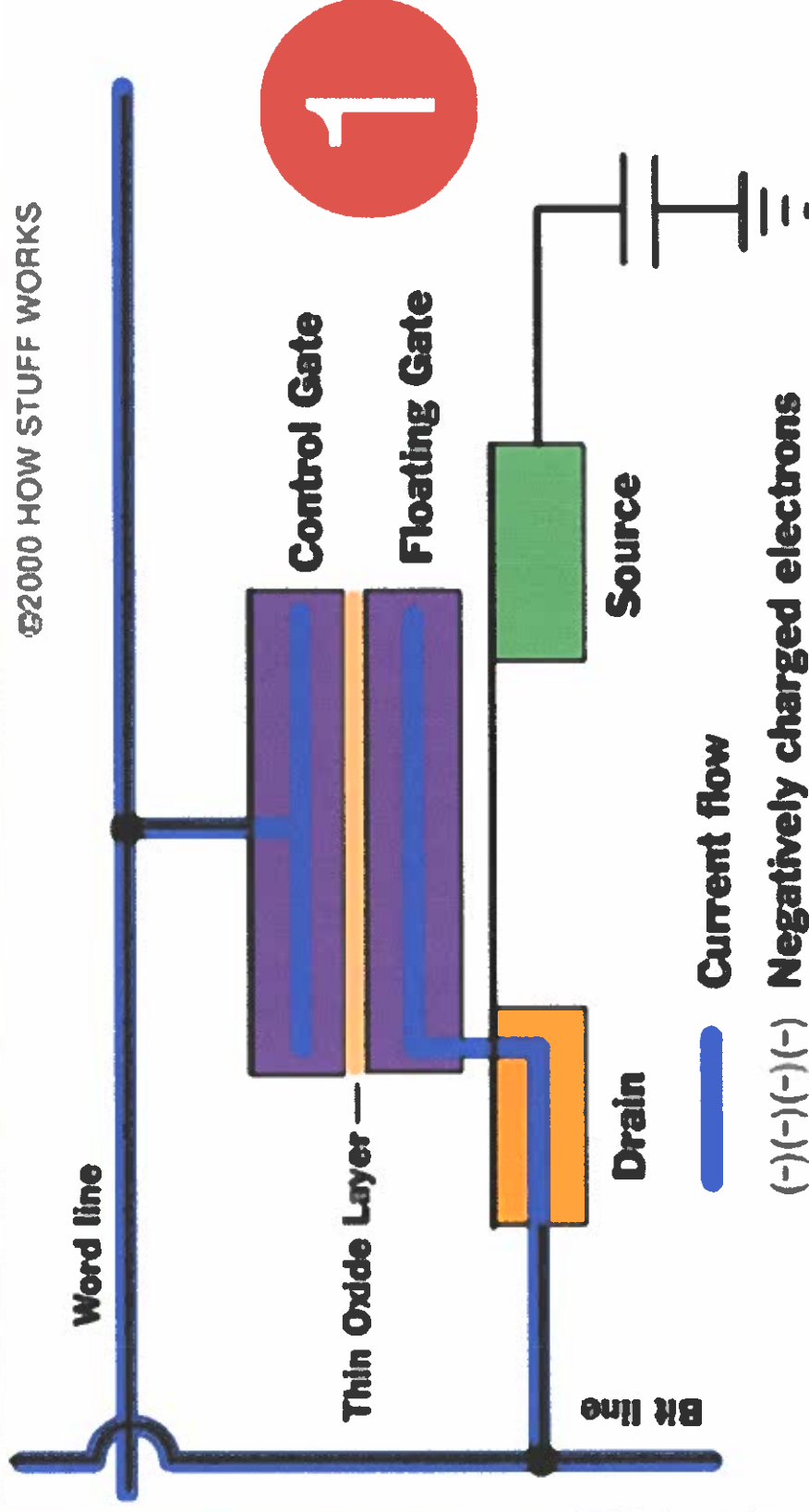
Brief History of ROM Types

- ▣ ROM: Read Only Memory
 - Like a CD ROM
- ▣ PROM: Programmable Read Only Memory
 - Like a Writable CD ROM
- ▣ EPROM: Erasable Programmable Read Only memory
- ▣ EEPROM: Electronically Erasable Programmable Read Only Memory
- ▣ Flash Memory: Sorta like a cross between PROM and EEPROM

Flash Memory - 0



Flash Memory - 1



Performance

- ▣ Random Access Times
 - Good USB Drive < 1ms
 - Hard Drive ~ 8ms
- ▣ Read Times
 - Good USB Drive ~ 20 MB/s
 - Hard Drive ~ 60 MS/s
- ▣ Write Times
 - Good USB Drive ~ 14 MB/s
 - Hard Drive ~ 60 MB/s

Performance

- ▣ USB Thumb Drives have slower transfer rates than hard drives, but have MUCH faster seek times
- ▣ Windows Vista and Windows 7 can use thumb drives in the cache hierarchy.
- ▣ Most beneficial when memory is low, and storing small frequently accessed pages, such as the page file

Performance Results

▣ Tests...

- First of all we installed Windows Vista Enterprise from scratch and added all required drivers. As we finished the installation we set a system restore point, because we recreated this clean system state (with no impact of SuperFetch learning from our testing activity) for every complete set of benchmarks. We used Microsoft Outlook with a 1.5 GB PST file, the Internet Explorer 7 with a locally stored website and the OpenOffice Writer with a huge press release document as our applications to test SuperFetch and ReadyBoost. Here is our testing sequence:

How It Works

- When a suitable USB drive is detected, the user is asked if they want to use ReadyBoost and how much storage to dedicate to it.
- Ecache.sys device driver intercepts all read/writes and copies them to the cache file on the thumb drive
- ReadyBoost compresses and encrypts the data for speed and safety
 - Advanced Encryption Standard (AES) encryption with a randomly generated per-boot session key
- When ReadyBoost sees random reads that the cache can satisfy, it utilizes it. Sequential reads are passed to the hard drive

Performance Results continued...

- Boot Windows Vista

- Wait 2 minutes

- **First Run**

- Start Internet Explorer : Wait 5 min (15 minutes if ReadyBoost USB 2.0 Flash device is attached for the first time) : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer : Wait 2 minutes

- **Second Run**

- Start Internet Explorer : Wait 2 minutes : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer : Wait 2 minutes

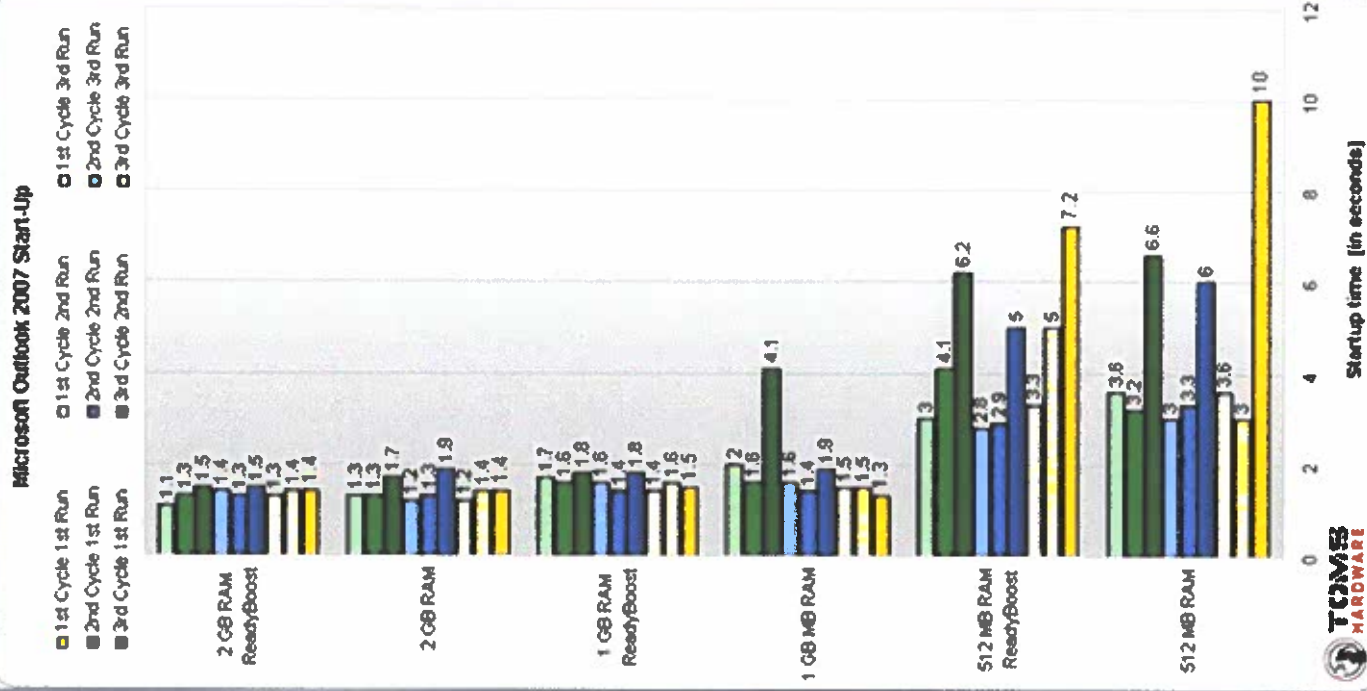
- **Third Run**

- Start Internet Explorer : Wait 2 minutes : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer :

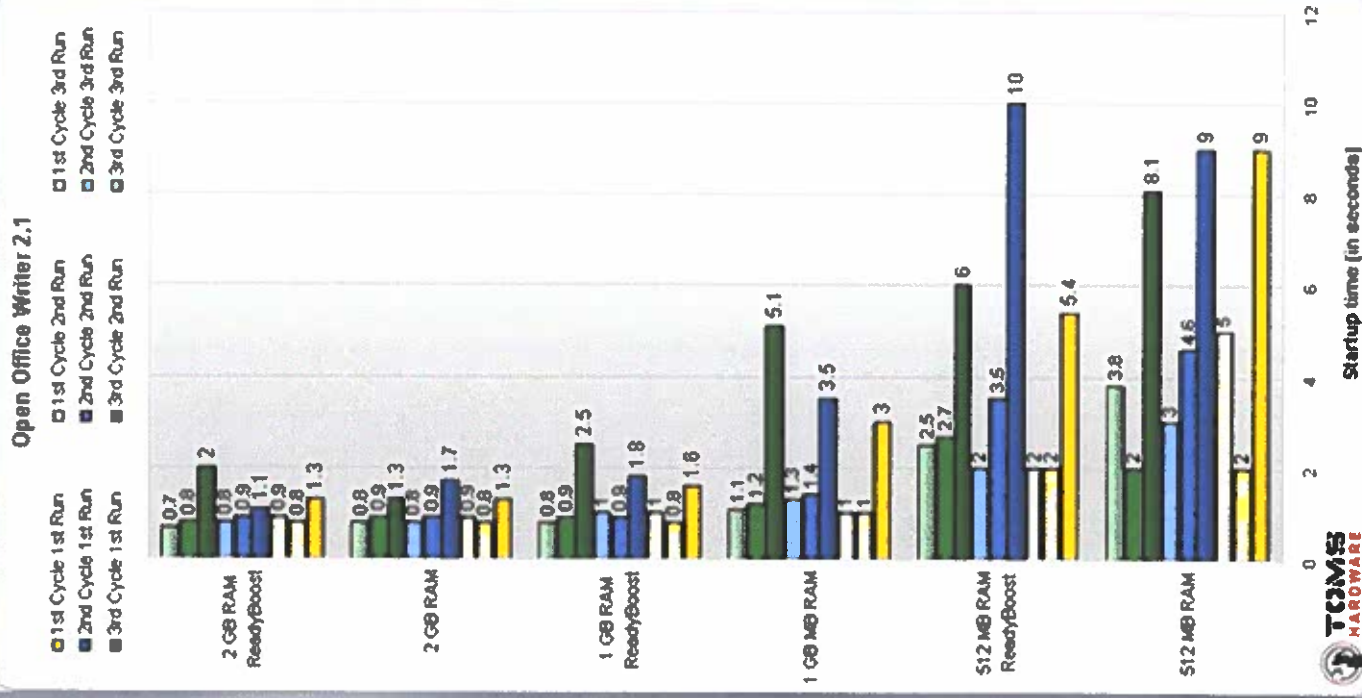
- Reboot and start 2nd or 3rd cycle

- We measured the startup time for Microsoft Office 2007 and the OpenOffice Writer 2.1

Performance Results continued...



Performance Results continued...



Conclusions

- ▣ ReadyBoost and SuperFetch can offer significant performance
- ▣ ReadyBoost can dramatically improve performance on low memory systems
 - Still some small gains after memory > 1gb
 - Recommendation: If you have a fast USB thumb drive, might as well use it.
- ▣ Superfetch does improve responsiveness if it can predict your behavior

WINDOWS VIRTUAL PC

Dave Coons

COIN 431

Virtual PC History

- ▣ Microsoft's line of PC Virtualization software
- ▣ Originally developed by Connectix
- ▣ Purchased by Microsoft and developed into Microsoft Virtual PC
- ▣ Released as a free product in 2006
- ▣ Upgraded in 2007 as Virtual PC 2007

Source: http://blogs.msdn.com/virtual_pc_guy/archive/2006/07/12/662535.aspx
<http://www.microsoft.com/presspass/press/2003/feb03/02-19PartitionPR.msp>

Added Features of Windows Virtual PC

- ▣ USB Support
- ▣ Seamless application publishing and launching
- ▣ Support for multi threads
 - Users will be able to run multiple virtual machines concurrently, each running in its own thread, which improves stability and performance.
- ▣ Clipboard sharing
- ▣ Printer redirection
 - Ability to redirect printing from guest machine to printers attached on host machine.

Windows Virtual PC

- ▣ More stringent requirements
- ▣ Unlike Microsoft Virtual PC, Windows Virtual PC requires hardware virtualization support
- ▣ Unlike Microsoft Virtual PC, Windows Virtual PC only runs on Windows 7
 - Microsoft Virtual PC 2007 'requires' Windows Vista Business, Windows Vista Enterprise, Windows Vista Ultimate, Windows XP Professional, Windows XP Tablet PC Edition, Windows Vista Service Pack 1 (SP1) (Enterprise, Business, Ultimate), and Windows XP Service Pack 3 (SP3).
 - Microsoft Virtual PC 2007 officially does not support home editions of XP and Vista, but the software will still run on those systems.

Source: <http://www.microsoft.com/windows/virtual-pc/support/virtual-pc-2007.aspx>

<http://www.microsoft.com/windows/virtual-pc/support/requirements.aspx>

Added Features of Windows Virtual PC

- ▣ Smart card redirection
- ▣ Drive sharing
- ▣ Integration with Windows Explorer
- ▣ Known folder integration between host and guest
- ▣ Support for higher resolutions
 - The allowable resolution for virtual machines has been extended up to 2048x1920 from 1600x1200.
- ▣ Integration with Windows XP Mode

Windows XP Mode

- ▣ Another free package from Microsoft
- ▣ Special Virtual Machine for Windows Virtual PC
 - Includes a free licensed copy of Windows XP Professional
- ▣ Seamlessly integrates with Windows 7
 - Allows software installed on Windows Virtual PC to be launched directly from Windows 7 Start Menu
- ▣ Implemented with Remote Desktop Protocol

Performance

- ▣ Running Passmark's Performance Test 7.0
- ▣ Compare Windows Virtual PC with VMware Player 3.0.0 build-203739
- ▣ Tests run on a Asus G50VT-X1
 - Host OS Windows 7 64 Enterprise Edition
 - 4Gb Ram
 - Intel P8400 @ 2.26GHz
 - nVidia GeForce 9800m GS w/ 512Gb 800MHz DDR3

VMware

- ▣ Principle competition to Microsoft's virtualization technology
- ▣ VMware player also offered free
- ▣ Also simulates a 'standard' pc
 - No exotic video or audio hardware
 - Aim is to have a virtual PC that can run on a variety of host machines, so the LCD is desired

Conclusion

- ▣ Windows Virtual PC has better integration than past incarnations of Virtual PC
- ▣ Windows Virtual PC is slower and less configurable than its competition
- ▣ Windows XP mode is an effective solution for businesses what wish to run legacy software not compatible with Windows 7
 - Performance hit is large, but legacy business software probably does not require modern performance levels
 - No chance of running legacy games

